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REVIEWS

ORAL HEALTH AND CARDIOVASCULAR DISEASES

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ABSTRACT

According to the World Health Organization (WHO), approximately 16% of the population's mortality is due to cardiovascular diseases worldwide, coronary artery disease being the predominant type. Cardiovascular diseases create social, economic and medical issues in many countries, which requires the urgent development of primary and secondary prevention programs. The main risk factors for coronary artery disease are well-known and the leading causal relationship is well-studied. Another leading healthcare problem worldwide is oral health. There is a number of studies that investigate the potential relationship between periodontitis and cardiovascular diseases. Periodontitis affects a large portion of the population, especially adults over 35 years. Worldwide, over 70% of the adults show some form of periodontal disease. When examining cardiovascular disease and periodontitis, similarities are found in the etiology, the common risk factors, as well as in the pathogenic mechanism. Chronic inflammation is a basis for both.

Keywords: *cardiovascular disease, atherosclerosis, periodontitis, Porphyromonas gingivalis, risk factors, inflammation, arterial hypertension, heart failure*

INTRODUCTION

Cardiovascular diseases (CVDs) are the leading cause of death in the world. The most common among them are coronary heart disease, arterial hypertension, heart failure, and stroke. According to WHO, in 2019, 17.9 million people died from CVDs, 32% of the total mortality worldwide. Cardiovascular diseases overtake cancer, which occupies second place as the most common cause of death with a percentage prevalence of 21.6% of total mortality. Nearly 85% of deaths are due to acute forms of atherosclerosis, namely stroke and myocardial infarction. In Europe, the data is similar, over 80% of deaths are due to heart attack or stroke [1]. In Bulgaria, over 60% of total mortality is due to CVDs, placing it among the countries with the highest mortality rate, about three times higher than the average [2].

Risk factors for CVD are modifiable and non-modifiable. They are well studied, and the causal relationship for the occurrence of cardiovascular disease is known. Modifiable risk factors are high blood pressure, dyslipidemia, smoking, unhealthy diet, intake of foods rich in saturated fats, sugars and salt, alcohol dependence, low physical activity, obesity, exposure to chronic stress. Non-modifiable risk factors include gender, age, ethnicity, genetic predisposition, and heredity.

A number of acute and chronic diseases, examined in numerous clinical studies, such as rheumatoid arthritis, psoriasis, hypothyroidism, type 2 diabetes mellitus, and chronic obstructive pulmonary

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disease (COPD), prove to be associated with the occurrence of cardiovascular diseases. The relationship between periodontitis and CVDs is studied not so well. Periodontitis is an inflammatory disease of the tooth-supporting apparatus, including the gums, periodontal ligament, cementum, and alveolar bone, that leads to tooth loss. It often goes unnoticed until symptoms such as pain, bleeding gums, bad breath, gum recession, loose teeth, and tooth loss occur in the more severe stages of periodontitis. The main cause is the accumulation of bacterial biofilm and the maintenance of chronic inflammation. A pre-existing condition of periodontitis is gingivitis, an inflammatory condition of the gums, which, if left untreated, is a major risk factor for the occurrence of periodontitis [3].

Oral health is a leading global goal and leads to the development of preventive and treatment strategies in a number of countries, due to the high incidence of dental diseases with systemic manifestations. According to WHO data, in 2021, about 951 million people suffered from periodontal diseases. In the same year nearly 80 million new cases were observed. The severe form of periodontitis affects over 1 billion people, with the highest percentage observed in South Asia. Currently, there are no current statistics on patients affected by periodontitis in Bulgaria. Forecasts show that by 2050, cases of severe periodontitis will globally reach 1.56 billion people, about 44% more compared with 2021 [4,5].

AIM

The aim of this article is to review current scientific evidence from studies on the potential link between periodontitis and cardiovascular disease.

MATERIALS AND METHODS

This article is based on a search, which was conducted across various databases like PubMed, Scopus, Science Direct, Google Scholar to gather the necessary information. The keywords used were: *cardiovascular diseases*, *atherosclerosis*, *periodontitis*, *Porphyromonas gingivalis*, *risk factors*, *inflammation*, *arterial hypertensio*n, *heart failur*e.

RESULTS

Several studies have attempted to establish a causal relationship between periodontitis and CVDs. Increasing evidence from multiple clinical trials suggests that chronic inflammation could potentially increase the risk of development and progression of CVDs [6]. By entering the bloodstream through direct invasion or through immune-mediated mechanisms, periodontal pathogens and their products can induce progression and change the proatherogenic and proinflammatory effects of endothelial cells and potentiate endothelial dysfunction, which underlies the atherosclerotic process [7]. Another emerging hypothesis suggests that bacteria that cause periodontitis influence and alter the gut or oral microbiome and may influence the inflammatory process in chronic diseases, including atherosclerosis [8]. Chronic inflammation in periodontitis increases the level of cytokines such as IL-1, IL-6, TNF-α, CRP, and fibrinogen, which accelerate atherogenesis and coagulation [9]. Oxidative stress, activation of growth factors, and subsequent remodeling of the vascular wall and cardiac tissue lead to potential consequences such as chronic heart failure and arrhythmias.

The most common oral pathogens are gram-negative, mainly anaerobes and some spirochetes: *Porphyromonas gingivalis*, *Aggregatibacter actinomycetemcomitans*, *Treponema denticola,* and *Fusobacterium nucleatum* [10]. *Porphyromonas gingivalis* is associated with severe forms of periodontitis, as is *Treponema denticola*, which is from the so-called red complex described by Dr. Sokransky, who defined these bacteria as key pathogens for the development of periodontitis and gingivitis [11–13]. The presence of these pathogens is associated with dysbiotic changes in the oral microbiome, which can lead to increased inflammation and tissue destruction [14–16]. Activation of platelet aggregation by oral bacteria, often *Streptococcus mutans* and *Streptococcus sanguinis*, can lead to the secretion of proinflammatory cytokines by platelets themselves, leading to inflammation, atherogenesis, and thrombogenesis [17].

Periodontitis and Atherosclerosis

In their study Yingle Wu et al. investigated the relationship between *Porphyromonas gingivalis*, systemic inflammation, and acute myocardial infarction. The subjects were 382 patients with acute myocardial infarction and 78 controls without CHD. Oral cell samples and blood samples were separated for genomic and cell-free DNA detection of *P. gingivalis* by quantitative PCR analysis. The results

showed that both local oral colonization and systemic colonization were associated with increased systemic inflammatory activity. An almost twofold increase in the risk for acute myocardial infarction was observed in the presence of *Porphyromonas gingivalis* in genomic DNA, and a correlation with more severe coronary lesions was also found [18].

Another meta-analysis by Quan Shi et al. confirmed that patients with acute myocardial infarction had worse periodontal status—more bacterial plaque, deeper periodontal pockets and bleeding, and these indicators were associated with inflammatory markers such as high-sensitivity CRP, troponin I, and natriuretic peptide.

In a Spanish cross-sectional study by the University of Granada, the authors studied 112 patients who had had acute myocardial infarction by monitoring the values of troponin I and myoglobin, which are biomarkers of myocardial necrosis, and also assessed the severity of periodontitis. They found a significant correlation between the severity and size of the infarction and the severity of periodontitis. The greater extent of periodontitis measured by the Arbes index correlated significantly with high troponin levels ($p < 0.02$). Severity was also associated with troponin I ($p = 0.04$) [19].

The Swedish Parokrank study was conducted between May 2010 and February 2014 in 17 Swedish hospitals, including 805 patients with acute myocardial infarction aged <75 years and 805 controls matched for age and sex. Assessment of periodontal status was based on panoramic radiographs, dividing patients into three groups: healthy ≥80% residual bone, very weak-moderate (66–79%) and severe (<66%). Moderate to severe periodontitis was observed in 43% of patients with myocardial infarction, compared to 33% of healthy controls. A total of 57% of patients with myocardial infarction had no periodontal disease, compared to 67% of the healthy controls. This shows that patients with infarction have significantly more severe forms of periodontitis ($p < 0.001$). Furthermore, the results indicate that periodontitis increases the risk of a first myocardial infarction. This means that people with moderate or severe periodontitis have a 28% higher risk of a first heart attack, even after accounting for traditional risk factors for coronary heart disease, such as smoking, type 2 diabetes, etc. The risk is even greater in younger patients and women <65 years of age. Women <65 years of age have a 3.72 times higher risk of a first heart attack. In women <60 years of age the risk is 5.26 times bigger. This is a red light that women with severe periodontitis are a particularly vulnerable group. Again, mechanisms such as chronic inflammation, invasion into the bloodstream and the occurrence of endothelial dysfunction and increased platelet activity are discussed [20].

There are also numerous studies that examine the relationship between periodontitis treatment and the impact on endothelial function.

Periodontitis and Arterial Hypertension

The association between periodontitis and hypertension has been well-documented in the scientific literature in recent years. A growing amount of research supports the hypothesis that chronic inflammation caused by periodontal pathogens may play a role in the development or exacerbation of arterial hypertension. Hypertension is a chronic disease characterized by persistently elevated blood pressure, which increases the risk of stroke, myocardial infarction, and renal failure. As it has been discussed, chronic inflammation in periodontitis releases proinflammatory mediators and cytokines. These molecules increase the risk of endothelial dysfunction, leading to vascular resistance, which is also a key mechanism for the development of arterial hypertension [21]. Disrupted oral microbiome by periodontal pathogens can reduce local nitrate production, also disrupting the production of nitric oxide, which normally maintains vascular vasodilation [22]. Meta-analyses and cohort studies have shown that patients with periodontitis have a higher risk of developing hypertension and that patients undergoing periodontal treatment have a tendency to decrease their blood pressure values. Czesnikiewicz-Guzik et al. published in 2019 in the European Heart Journal a clinical trial in which patients with arterial hypertension and periodontitis were randomized to receive non-surgical intensive therapy (subgingival and supragingival scaling with chlorhexidine, or control periodontal treatment, which included only supragingival scaling). As a main parameter for evaluating results, they monitored 24-hour systolic and diastolic blood pressure for 2 months, re-

porting a significant improvement in systolic blood pressure values, as well as in periodontal status. Systolic blood pressure fell by an average of 11.2 mmHg and diastolic by 8.4 mmHg 2 months after therapy. They used genetic (natural) randomization, examining SNPs genetic variants associated with periodontitis (SIGLEC5 and other genes). and data from about 750,000 participants from the UK-Biobank and the International BP GWAS consortium. The analysis showed a significant causal relationship between the genetic SNPs and blood pressure values. Individuals with a high genetic predisposition to periodontitis had higher systolic and diastolic blood pressure values. The combination of genetic data and remission therapy provides strong evidence for the causal role of periodontitis in the development and exacerbation of hypertension. The importance of this study is related to the potential beneficial effect of periodontal therapy on the maintenance of good cardiovascular health [23]. Collaboration between dentists, cardiologists, and general practitioners is required in the treatment of chronic inflammatory conditions.

Periodontitis and Heart Failure

The relationship between periodontitis and heart failure is a subject of growing scientific interest. Although the topic is less studied than arterial hypertension and atherosclerosis, there is convincing evidence that chronic periodontitis may contribute to the development and progression of heart failure, particularly through inflammatory and immune mechanisms.

According to the European Society of Cardiology, heart failure is defined as a clinical syndrome characterized by symptoms of shortness of breath, fatigue, swelling of the lower extremities, accompanied by signs of increased jugular venous pressure and moist pulmonary rales. It results from structural or functional abnormalities of the heart in which, despite adequate ventricular filling, cardiac output is reduced and the heart is unable to pump the necessary amount of blood to meet the metabolic needs of the tissues [24].

The importance of chronic inflammation and the production of inflammatory markers, which are directly related to endothelial dysfunction, cardiomyocyte damage, myocardial fibrosis, deterioration of left ventricular ejection fraction and the development of a clinical picture of heart failure, is again discussed in the scientific literature. Another hypothesis suggests that bacteria in the oral cavity, e.g., *P. gingivalis*, which is a key pathogen in severe forms of periodontitis, during various activities such as brushing teeth or dental manipulations, can enter the bloodstream, cause bacteremia and be deposited in the endocardium and myocardium, causes an educational process [25]. A particularly vulnerable group are the patients with underlying structural heart disease, who need antibiotic prophylaxis when performing dental manipulations [26].

The first large observational cross-sectional study on the topic was conducted in the period 1990–1994, but published finally in 2021 in the Journal of Clinical Periodontology. The study included 13,202 participants based on data from NHANES III-USA. The diagnosis of periodontitis was made based on clinical signs: loss of attachment, pocket depth, presence of bleeding. The presence of heart failure was based on questionnaires and accompanying medical documentation, which confirmed the diagnosis. After excluding other risk factors known for heart failure, such as type 2 diabetes, smoking, age, gender, type 2 diabetes, BMI, the correlation between severe periodontitis and heart failure remained high ($p < 0.01$). Limitations of the study included the crossover design, which did not prove a direct causal relationship. The diagnosis of heart failure was based only on anamnestic data and medical documentation, but not on direct examination of laboratory biomarkers and echocardiographic parameters to assess left ventricular systolic and diastolic function [27]. In contrast, the German study, Hamburg City Health Study, started in 2016. A total of 10,000 participants aged 45–74 years were studied and echocardiographic and medical criteria were included for the diagnosis of heart failure. Patients were divided into three groups: heart failure with reduced ejection fraction, heart failure with mildly reduced ejection fraction and heart failure with preserved ejection fraction. A strong correlation was observed between severe periodontitis and heart failure with reduced and mildly reduced ejection fraction. Severe periodontitis was associated with a 2.17-fold higher risk of heart failure with mildly reduced ejection fraction and a 1.62-fold higher risk of heart failure with reduced ejection fraction. The limitations of this study were related to

its design, a cross-over, which did not show causality, but only association, and there was no long-term follow-up to assess the progression of heart failure [28].

DISCUSSION

The accumulated scientific evidence suggests a consistent association between periodontitis and CVDs, particularly atherosclerosis, arterial hypertension, and heart failure. Although the majority of studies demonstrate correlation, the causal relationship has not been definitively established, largely due to the predominance of cross-sectional and observational designs, as well as the presence of multiple confounding variables such as smoking, diabetes, and socioeconomic status [6,7,18,32].

Nevertheless, mechanistic hypothesis provides a plausible explanation for the link. Chronic systemic inflammation, bacteremia induced by oral pathogens, and immune-mediated endothelial dysfunction represent key shared pathogenic pathways [9,14,31]. Periodontal bacteria such as *Porphyromonas gingivalis* have been shown to influence endothelial cell activity, promote platelet aggregation and contribute to vascular remodeling [10–13]. Elevated inflammatory mediators (IL-6, TNF-α, CRP) in patients with periodontitis are consistent with biomarkers associated with increased cardiovascular risk [9,14,34].

Importantly, interventional studies evaluating periodontal treatment have reported improvements in surrogate cardiovascular outcomes, such as reductions in systemic inflammation and blood pressure, although data remain limited and heterogeneous [21–23]. Future large-scale, longitudinal, and randomized controlled trials are needed to clarify the extent to which periodontal therapy can reduce cardiovascular morbidity and mortality [3,4,28,31].

A multidisciplinary approach involving dentists, cardiologist, and general practitioners appears essential for the early identification of patients at risk and for implementing preventive strategies. Integrating oral health into cardiovascular risk assessment could represent an important step toward precision medicine and holistic patient care [2,23,28,33].

CONCLUSION

The relationship between oral health and CVDs is increasingly supported by epidemiological, clinical, and mechanistic evidence. While a direct causal link has not yet been definitively proven, the overlap in risk factors and shared inflammatory pathways underscores the importance of maintaining good periodontal health as part of overall cardiovascular prevention [3,5,32,33].

Population-based studies highlight that improved oral hygiene practices, such as frequent tooth brushing and regular dental visits, are associated with a measurable reduction in major adverse cardiovascular events [29,30]. This suggest that oral hygiene may serve as a simple, cost-effective and widely applicable strategy to mitigate cardiovascular risk at the population level.

In conclusion, maintaining optimal periodontal health should be considered an integral component of CVD prevention. Further well-designed interventional studies are warranted to confirm causality and to determine the precise clinical benefits of periodontal treatment in reducing cardiovascular morbidity and mortality. Collaborative efforts between dental and medical professionals will be crucial to advancing patient care and improving long-term health outcomes [1–5,23,28,33].

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